

TUTORIAL TASKS:11-20

11.AI Search Comparison

Description:

The AI Search Comparison system helps users compare answers generated by different AI tools for the same question. It is used to evaluate which AI tool provides more accurate, clear, detailed, and useful responses.

Features

- Search query input system
- Multiple AI tool comparison
- Answer accuracy checking
- Response clarity evaluation
- Reference and source comparison
- Comparison table generation
- Final tool recommendation
- User-friendly interface

AI Tools Used

- **ChatGPT** – Detailed explanations and structured answers
- **Google Gemini** – Updated search-based responses
- **Perplexity AI** – Answers with source references
- **Microsoft Copilot** – Professional and concise responses
- **Google Search** – Website-based information verification

Workflow

1. User enters a common question or topic.
2. The same question is searched in different AI tools.
3. Answers from all tools are collected.
4. Responses are compared based on accuracy, clarity, depth, and references.
5. A comparison table is prepared.
6. The best-performing AI tool is identified.
7. Final observations are displayed to the user.

Benefits

- Improves research quality
- Helps select the best AI tool
- Supports fact-checking
- Builds critical thinking
- Reduces dependency on a single source

1 AI Search Comparison

What is Artificial Intelligence?



Tool	Summary
 ChatGPT	AI is the simulation of human intelligence in machines.
 Google Gemini	AI refers to machines performing tasks that normally require human intelligence.
 Perplexity AI	AI is a field of computer science focused on building smart systems.
 Microsoft Copilot	AI enables systems to learn, reason and solve problems.
 Google Search	AI is the intelligence shown by machines that mimics humans.

Best Overall: Perplexity AI (For references)
ChatGPT (For detailed explanations)

12.AI Note Summarization

Description:

The AI Note Summarization system helps users convert long notes, lecture content, PDFs, or study materials into short, clear, and meaningful summaries. It is useful for quick revision and better understanding.

Features

- Upload or paste notes
- AI-based summarization
- Key point extraction
- Important definitions identification
- Flashcard generation
- Question-and-answer creation
- Exam revision notes
- Downloadable summarized output

AI Tools Used

- **ChatGPT** – Summarizing notes and creating key points
- **Claude** – Long document summarization
- **Notion AI** – Organizing summarized notes
- **Microsoft Copilot** – Summarizing Word documents
- **PDF AI Tools** – Extracting content from PDF files

Workflow

1. User uploads or pastes long notes.
2. The AI reads and analyzes the content.
3. Important points and repeated information are identified.
4. Unnecessary content is removed.
5. AI creates a short summary with headings and bullet points.
6. Key terms, definitions, and questions are generated.
7. Final summarized notes are displayed to the user.

Benefits

- Saves study time
- Helps in quick revision
- Improves understanding
- Converts lengthy content into simple notes
- Useful for exam preparation

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AI Note Summarization



Original Notes

Long lecture notes, articles, or study material



AI Generated Summary

- Key Points
- Important Definitions
- Examples
- Questions & Answers
- Flashcards



Short, Clear & Easy to Revise!



13.AI Career Roadmap Creation

Description:

The AI Career Roadmap Creation system helps users prepare a step-by-step career development plan based on their qualifications, skills, interests, and target job role.

Features

- Career goal input
- Skill gap analysis
- Personalized roadmap generation
- Course and certification suggestions
- Project recommendations
- Resume improvement tips
- Interview preparation plan
- Monthly progress tracker

AI Tools Used

- **ChatGPT** – Career planning and roadmap creation
- **Google Gemini** – Career research and learning resources
- **LinkedIn AI Tools** – Profile improvement and job search support
- **Resume Builder AI Tools** – Resume creation and optimization
- **Microsoft Copilot** – Professional document preparation

Workflow

1. User enters current qualification and career goal.
2. AI analyzes the required skills for the target role.
3. Skill gaps are identified.
4. AI creates a weekly or monthly career roadmap.
5. Courses, certifications, and projects are suggested.
6. Resume and LinkedIn improvement tips are generated.
7. Interview preparation steps are displayed to the user.

Benefits

- Gives clear career direction
- Helps in skill development
- Reduces career confusion
- Supports resume and interview preparation
- Improves job readiness

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AI Career Roadmap Creation

Target Role: Data Scientist

1	Learn Fundamentals Python, Statistics, SQL	
2	Data Analysis Pandas, NumPy, Excel	
3	Machine Learning Scikit-learn, ML Models	
4	Projects Build 3-5 Real Projects	
5	Certifications Courses & Certificates	
6	Job Preparation Resume, Interview, Portfolio	

Goal: Get a Data Scientist Job



14.AI Time Management Planner

Description:

The AI Time Management Planner helps users organize their daily, weekly, or monthly activities. It creates a structured timetable based on tasks, deadlines, priorities, and available time.

Features

- Daily task input
- Priority-based planning
- Smart timetable generation
- Deadline tracking
- Break and rest scheduling
- Study/work planner
- Reminder system
- Progress tracking dashboard

AI Tools Used

- **ChatGPT** – Timetable creation and task planning
- **Google Calendar** – Scheduling and reminders
- **Notion AI** – Task organization and planning
- **Todoist AI** – Task management and prioritization
- **ClickUp AI** – Workflow and productivity planning

Workflow

1. User enters daily tasks, deadlines, and available time.
2. AI analyzes task urgency and importance.
3. Tasks are arranged based on priority.
4. A daily or weekly timetable is generated.
5. Breaks, meals, study/work time, and rest are included.
6. Reminders are created for important tasks.
7. Progress is tracked and updated regularly.

Benefits

- Better time management
- Reduces stress
- Avoids last-minute work
- Improves discipline
- Helps complete tasks on time

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AI Time Management Planner



Daily Schedule



06:00 – 07:00		Exercise
07:00 – 08:00		Breakfast & Get Ready
08:00 – 10:00		Deep Work / Study
10:00 – 10:30		Break
10:30 – 01:00		Classes / Work
01:00 – 02:00		Lunch Break
02:00 – 04:00		Work / Study
04:00 – 04:30		Break
04:30 – 06:00		Assignments / Projects
06:00 – 07:00		Revision
07:00 – 08:00		Dinner
08:00 – 10:00		Reading / Learning
10:00 – 10:30		Plan for Tomorrow
10:30 PM		Sleep

15.AI Reading Assistant

Description:

The AI Reading Assistant helps users understand difficult reading materials such as textbooks, articles, research papers, and reports. It simplifies complex content and explains it in easy language.

Features

- Text or PDF input
- Simple explanation generation
- Difficult word meaning
- Key point extraction
- Summary creation
- Example generation
- Flashcard creation
- Question-and-answer preparation

AI Tools Used

- **ChatGPT** – Simplifying and explaining content
- **Claude** – Long article understanding
- **Perplexity AI** – Reference-supported explanations
- **PDF AI Readers** – Reading and extracting PDF content
- **Microsoft Copilot** – Document explanation and summarization

Workflow

1. User uploads or pastes reading material.
2. AI reads and analyzes the content.
3. Complex sentences and difficult words are identified.
4. AI explains the content in simple language.
5. Key points and important terms are extracted.
6. Questions, answers, and flashcards are generated.
7. Simplified reading notes are displayed to the user.

Benefits

- Makes difficult topics easy to understand
- Improves reading comprehension
- Helps in quick revision
- Supports self-learning
- Useful for academic reading

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AI Reading Assistant

Upload Text / PDF



AI Processing



Output

- Simple Explanation
- Key Points
- Difficult Words Meaning
- Summary
- Questions & Answers



16.AI Coding Assistant Exploration

Description:

The AI Coding Assistant helps users write, understand, debug, and improve programming code. It supports beginners and developers by providing code suggestions and explanations.

Features

- Code generation
- Code explanation
- Error detection
- Debugging support
- Syntax correction
- Code optimization
- Comment generation
- Multi-language programming support

AI Tools Used

- **ChatGPT** – Code writing, explanation, and debugging
- **GitHub Copilot** – Real-time code suggestions
- **Replit AI** – Online code generation and testing
- **Codeium** – Code completion support
- **Tabnine** – AI-based coding assistance

Workflow

1. User selects a coding task.
2. User enters the programming language and requirements.
3. AI generates the required code.
4. The code is tested in a coding editor.
5. Errors are copied and given back to AI.
6. AI identifies and corrects the errors.
7. Improved and final working code is generated.

Benefits

- Helps beginners learn coding
- Saves programming time
- Improves debugging skills
- Explains code clearly
- Supports faster software development

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AI Coding Assistant Exploration

```
1 def add(a, b):  
2     return a + b  
3  
4 num1 = 10  
5 num2 = 20  
6 result = add(num1, num2)  
7 print("Result:", result)  
8
```



AI Suggestion

```
# Function to add two numbers  
def add(a, b):  
    return a + b
```



AI helps: Code Generation, Explanation,
Error Fixing, Optimization

17.AI Productivity Workflow Mapping

Description:

The AI Productivity Workflow Mapping system helps users break down a task or process into clear steps. It creates an organized workflow from starting point to final completion.

Features

- Task input system
- Step-by-step workflow creation
- Checklist generation
- Timeline planning
- Priority identification
- Task board creation
- Progress monitoring
- Workflow improvement suggestions

AI Tools Used

- **ChatGPT** – Workflow planning and checklist creation
- **Notion AI** – Productivity templates and task boards
- **Trello AI** – Visual workflow management
- **Asana AI** – Project planning and task tracking
- **ClickUp AI** – Workflow automation and productivity tracking

Workflow

1. User enters a task or process.
2. AI identifies the main goal.
3. The task is divided into smaller steps.
4. Steps are arranged in proper order.
5. A checklist or workflow is created.
6. Deadlines and priorities are added.
7. Progress is tracked until completion.

Benefits

- Organizes work clearly
- Reduces confusion
- Improves task completion
- Saves time and effort
- Helps manage academic and professional work

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AI Productivity Workflow Mapping

Project Workflow



18.AI Knowledge Extraction Exercise

Description:

The AI Knowledge Extraction system helps users extract important information from large documents, articles, reports, or datasets. It identifies key facts, findings, conclusions, and useful points.

Features

- Document upload or text input
- Key information extraction
- Aim and objective identification
- Methodology extraction
- Result and finding extraction
- Limitation identification
- Table-based output
- Summary generation

AI Tools Used

- **ChatGPT** – Information extraction and organization
- **Claude** – Long document analysis
- **Perplexity AI** – Reference-supported extraction
- **PDF AI Tools** – Extracting data from PDFs
- **Elicit** – Research article analysis

Workflow

1. User uploads or pastes a document.
2. AI reads and analyzes the content.
3. Important information is identified.
4. Details such as aim, methods, results, and conclusion are extracted.
5. The extracted information is arranged in a table.
6. User reviews the extracted points.
7. Final structured knowledge output is displayed.

Benefits

- Saves reading time
- Helps in research work
- Improves information organization
- Supports literature review writing
- Extracts only useful and important content

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AI Knowledge
Extraction Exercise



Input: Research Paper / Document



AI Extracts	
Aim	To study the impact of AI in education.
Methodology	Survey & Data Analysis
Sample Size	200 Students
Results	AI tools improved learning by 65%.
Conclusion	AI enhances learning experience.
Limitations	Limited to one region.

Structured Knowledge in Seconds! ✨

19.AI Research Log Creation

Description:

The AI Research Log system helps users maintain a structured record of research articles, references, findings, and notes. It is useful for thesis writing, literature review, and academic projects.

Features

- Research topic input
- Article tracking table
- Summary generation
- Reference management
- Key findings recording
- Limitation tracking
- Citation status monitoring
- Research progress dashboard

AI Tools Used

- **ChatGPT** – Research log template and article summaries
- **Notion AI** – Research database management
- **Google Sheets** – Article tracking and sorting
- **Microsoft Excel** – Structured research log creation
- **Zotero / Mendeley** – Citation and reference management
- **Elicit** – Research article discovery and extraction

Workflow

1. User selects a research topic.
2. Relevant articles are collected.
3. AI creates a research log template.
4. Article details are entered into the log.
5. AI summarizes each article.
6. Key findings, limitations, and relevance are recorded.
7. The research log is updated regularly for writing and review.

Benefits

- Organizes research articles
- Helps in thesis and report writing
- Prevents missing important references
- Makes literature review easier
- Tracks research progress clearly

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AI Research Log Creation

Research Topic: AI in Education

Author & Year	Title	Methodology	Key Findings	Notes
Smith (2023)	Impact of AI Tools	Survey	AI improves engagement	Useful
Johnson (2022)	AI in Classroom	Case Study	Better learning outcomes	Important
Williams (2021)	AI Based Learning Systems	Experiment	AI increases productivity	Good

Summary

AI tools are effective in improving learning and productivity.



20.AI Learning Dashboard

Description:

The AI Learning Dashboard helps users track their learning progress for a subject or skill. It organizes topics, tasks, practice activities, revision dates, scores, and weak areas.

Features

- Subject or skill input
- Topic-wise learning plan
- Daily or weekly task tracker
- Practice activity planner
- Progress status dashboard
- Score tracking system
- Weak area identification
- Revision reminder system

AI Tools Used

- **ChatGPT** – Learning plan and progress tracker creation
- **Notion AI** – Learning dashboard and study organization
- **Google Sheets** – Progress tracking and charts
- **Microsoft Excel** – Learning records and score tracking
- **Trello / ClickUp** – Visual task tracking

Workflow

1. User selects a subject or skill to learn.
2. AI divides the subject into smaller topics.
3. A daily or weekly learning plan is created.
4. Practice tasks and revision dates are added.
5. User updates completion status and scores.
6. AI identifies weak areas based on progress.
7. Suggestions for improvement are displayed to the user.

Benefits

- Tracks learning progress
- Helps identify weak topics
- Improves study consistency
- Supports goal-based learning
- Makes learning organized and measurable

10 AI Learning Dashboard

Total Topics

24

Completed

16

In Progress

6

Pending

2

Progress Overview



Subjects

- Python 75%
- Data Structures 60%
- DBMS 70%
- AI/ML 50%

Upcoming Tasks

- Complete Python Functions
- Study Machine Learning Basics
- Practice Python Problems

Weak Areas

- AI Algorithms
- Statistics
- SQL Joins

